
Impact Statement

Optimizing RAG Document Retrieval for Agronomic Advice

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Positive Impact

The Agricultural Extension RAG project has the potential to improve access to factual, expert-validated agronomic information for farmers who currently lack reliable connectivity to the internet or to trained agriculturalists. By bridging this information gap, the system can help smallholder farmers make better-informed decisions, expand their activities more securely, and ultimately strengthen household food security. It also contributes to reducing the inequality that persists between rural farmers and those with privileged access to expertise and digital resources.

Beyond the immediate agricultural setting, the retrieval methods explored here—BM25, dense embeddings, re-rankers, and hybrid approaches—are transferable to other low-resource, domain-specific information retrieval scenarios such as community health advisory. The project democratizes access to agronomic knowledge often buried in PDF repositories, archives, or research databases and inaccessible to farmers and field agents. An effective retrieval system lowers this barrier regardless of a user's literacy level, proximity to an extension office, or internet access.

Risks and Limitations

Albeit the apparent benefit of this work, several aspects deserve keen attention.

The risk of *over-reliance* on retrieved advice is not negligible—for instance, recommending a pesticide regime designed for a different agroecological zone. A farmer who trusts the output could suffer crop damage or financial loss.

Equally, the quality and coverage of the underlying knowledge base set the ceiling of the system's usefulness. Existing biases in the corpus—such as over-representation of certain crops (e.g., maize, cocoa) or certain regions—and any erroneous or outdated information it contains may propagate into the system's recommendations, systematically under-serving farmers who grow less-studied crops or farm in less-documented areas.

Data Governance

The dataset used in this work consists of publicly available agricultural extension material and does not contain personally identifiable farmer data. However, in deployment scenarios where farmer queries are logged to improve the system over time, appropriate data governance measures must be established.



Future Work

Future work should:

- Involve smallholder farmers and extension agents in evaluating retrieval quality.
- Develop models that handle multilingual and code-switched queries.
- Establish protocols for reviewing, updating, and retiring documents.
- Release the evaluation framework and relevance judgements as an open benchmark, enabling broader contributions while maintaining scientific rigour.